

An orange circle containing the text "BDD 100K" in white, bold, sans-serif font.

**BDD
100K**

DATASCI 281 • FUNDAMENTALS OF COMPUTER VISION

Driving-Scene Classification on BDD100K

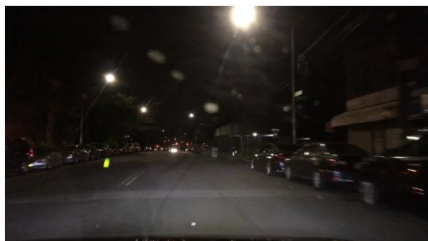
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THE PROBLEM

A single dash cam frame, six driving environments

BDD100K (Berkeley DeepDrive) — 100,000 keyframes at 1280×720 RGB, one frame sampled from the 10th second of each 40-second clip, across 50,000+ rides in New York, the SF Bay Area and other US regions. We predict the scene tag: the most visually discriminative of the three image-level attributes.



City Street



Highway



Parking Lot



Residential



Tunnel



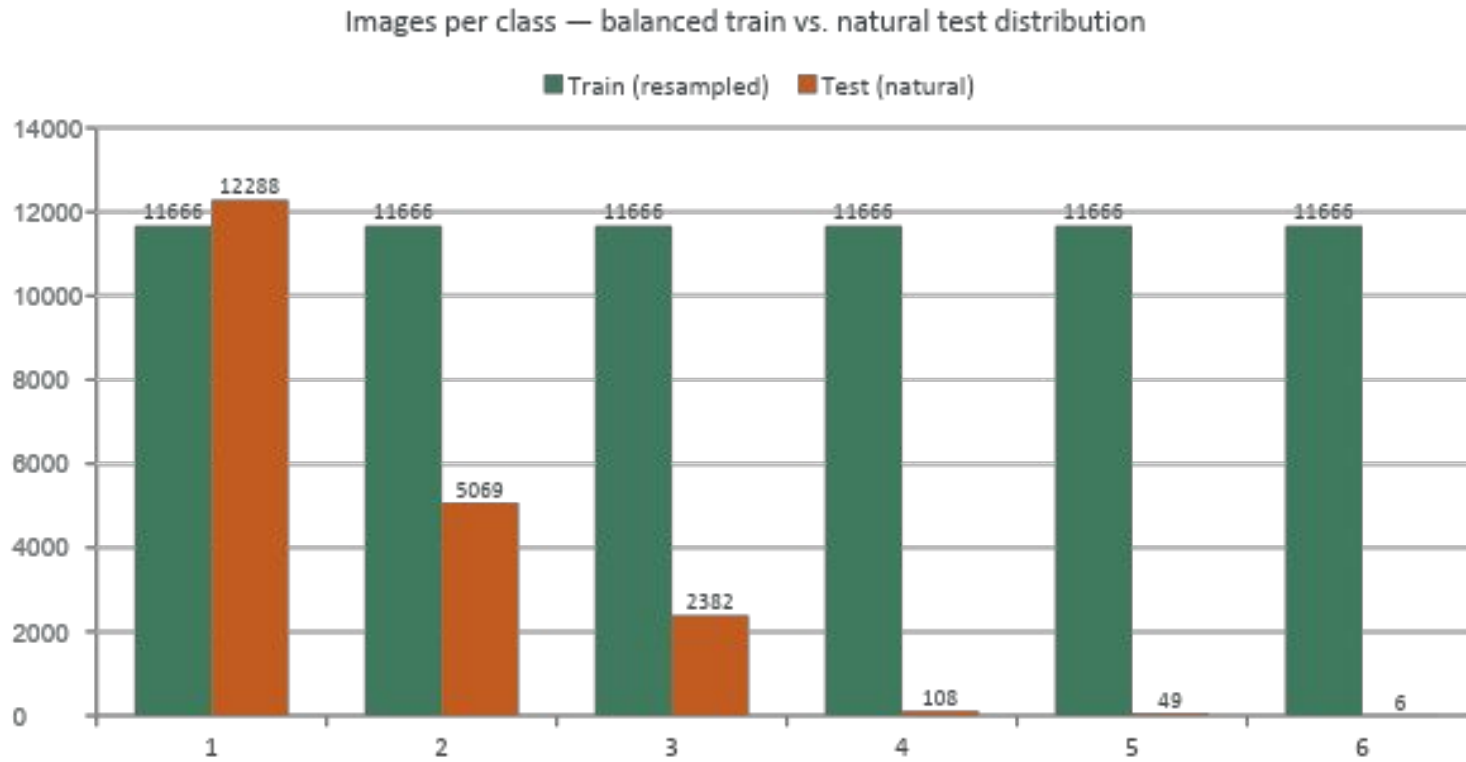
Gas Station

COMPLEXITIES

- 7 weather conditions — clear through snow and fog — change colour and contrast wholesale
- 4 times of day, so illumination varies dramatically inside every scene class
- 50,000+ distinct rides, so many road layouts and architectures per class
- Heavy foreground clutter — 2M+ annotated objects — differs even within one scene type
- Severe class imbalance among scene tags

Rebalancing the training set

Our proposal flagged severe class imbalance as a major concern. Rather than reweighting the loss, we rebuilt the training set: the official BDD100K train partition was resampled to exactly 11,666 images per class. Validation and test keep their natural distributions so the evaluation stays honest.

**69,996**

training images, 11,666 per class

9,947

validation images, natural imbalance

19,902

test images, natural imbalance

2,048×

imbalance in test — 12,288 city street vs 6 gas stations

Simple features chosen for global scene layout

Driving scenes are defined by a combination of color, texture, geometric structure, and spatial arrangement rather than any single object. Images were center-cropped to 1000×600 pixels and resized it to 250×150 pixels before extracting twelve feature families.

1 Color and Intensity

Features: RGB/HSV histograms, regional histograms, color-intensity

Captures overall color and brightness, while top/middle/bottom regions preserve rough vertical structure such as sky, buildings, and road.

2 Texture

Features: LBP, multi-scale LBP, GLCM

Measures local and multi-scale texture, including repeated patterns and gray-level relationships that may distinguish pavement, buildings, and tunnel surfaces.

3 Edges and Geometry

Features: HOG, edge summary, Hough Lines

Captures edge strength, gradient orientation, and straight-line directions associated with lane markings, buildings, highways, and tunnels

4 Spatial Layout

Feature: Coarse spatial grid

Divides the image into a 3×3 grid and summarizes local color, intensity, edges, and orientations within each region

All 12 handcrafted features were extracted first and backward elimination later removed HOG and region-wise HSV from the final handcrafted set.

Two Pretrained Embeddings as the complex feature

ResNet50

2,048 dims · 7.0 ms/image

ImageNet-pretrained convolutional network. We replaced the final classifier with an identity layer and used the pooled feature representation.

ViT-B/16

768 dims · 15.0 ms/image

ImageNet-pretrained vision transformer. We removed the classification head and used the final transformer embedding.

Motivation

- ImageNet pretraining provides higher-level representations that can respond to structures common in driving scenes, such as roads, buildings, vehicles, vegetation, and sky.
- These representations are less dependent on exact pixel colors than raw histograms, which may help across changes in lighting and weather.
- Both backbones were kept frozen, allowing the handcrafted and pretrained features to be compared using the same downstream classifier.

EXTRACTION INVENTORY

14

feature families per image

99,845

images across all three splits

7,948

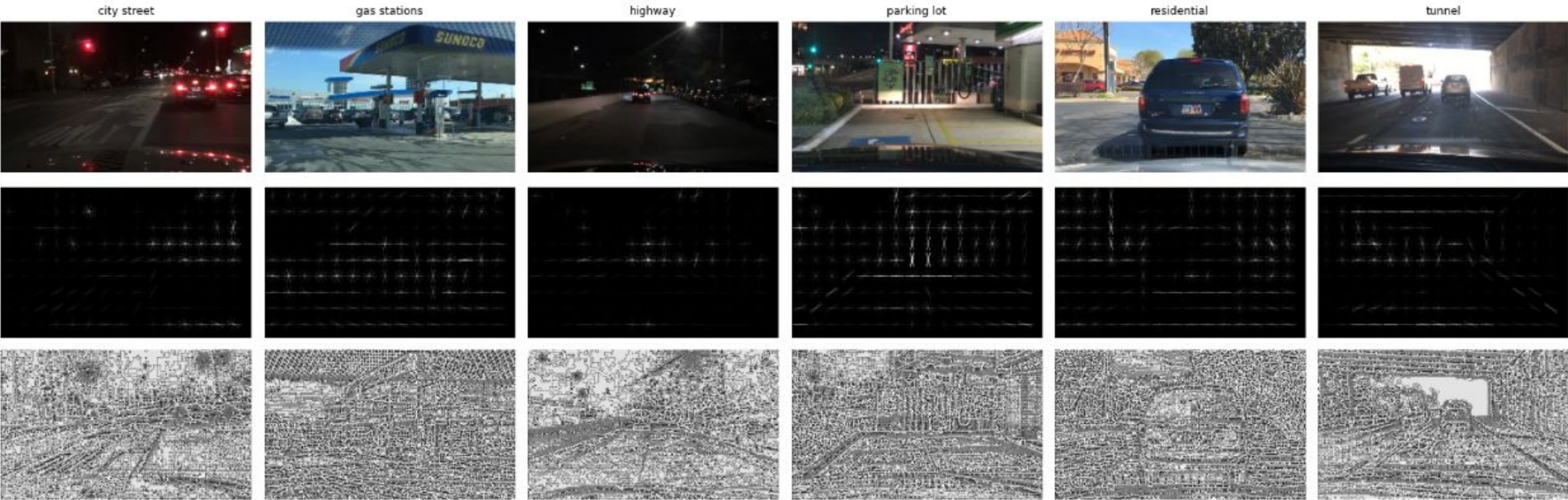
raw concatenated dimensions

4h 04m

total extraction wall time

Final sets: Set A = 956 dims · Set B = 3,772 dims

FEATURE ILLUSTRATION · 1 OF 2



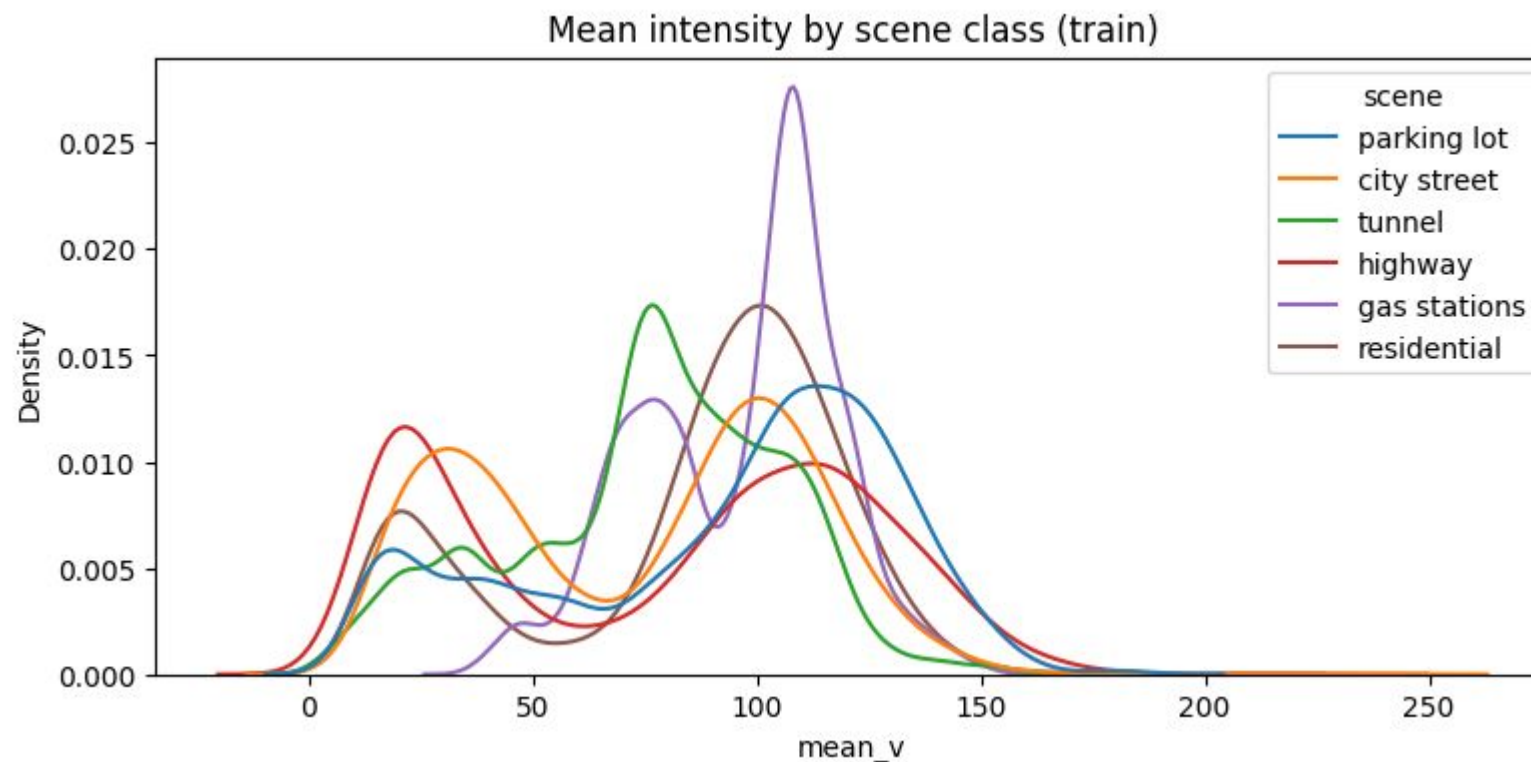
Original: city street

Grayscale

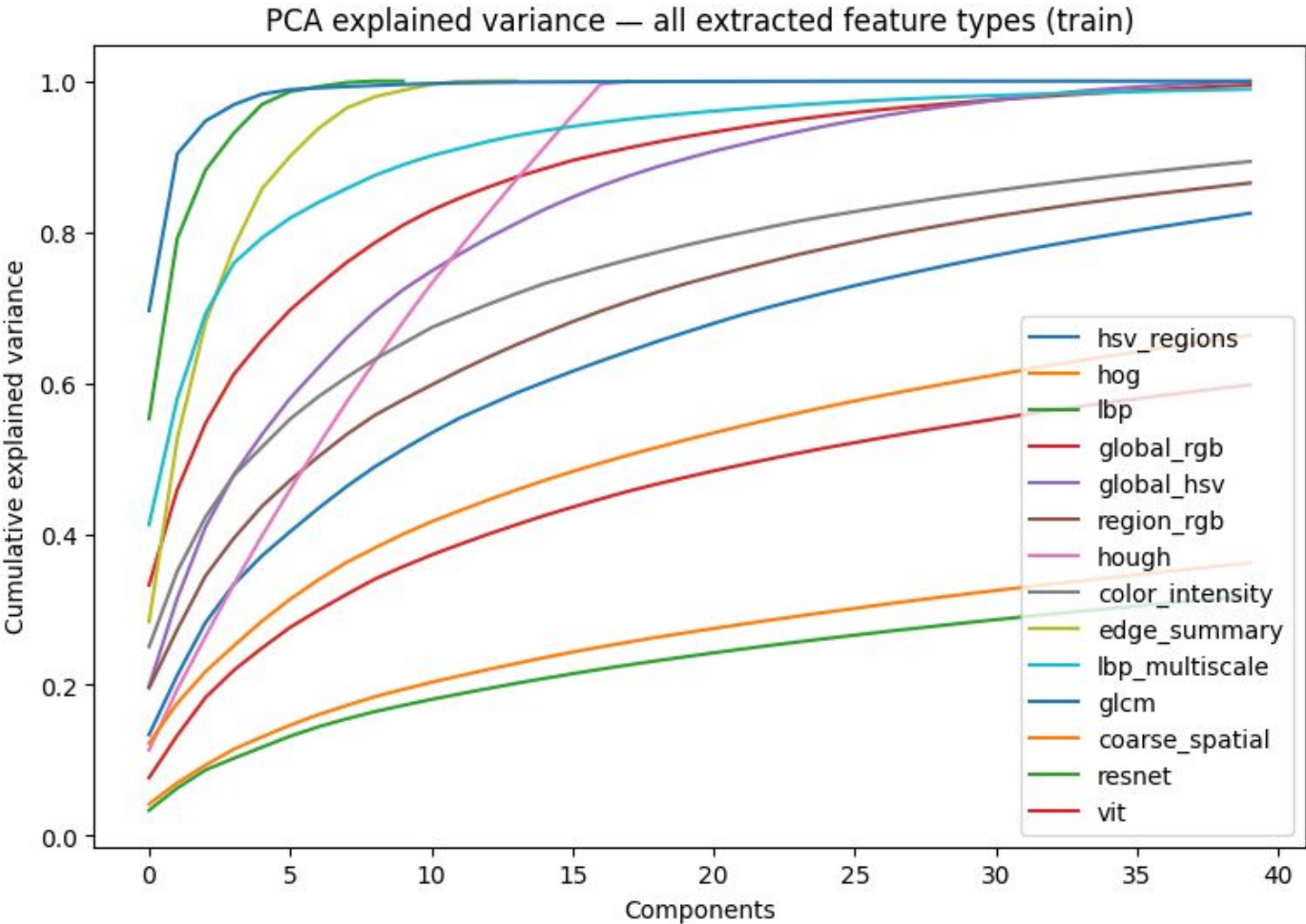
Sobel edge magnitude



Dataset variation: mean brightness by class



PCA - Cumulative Explained Variance

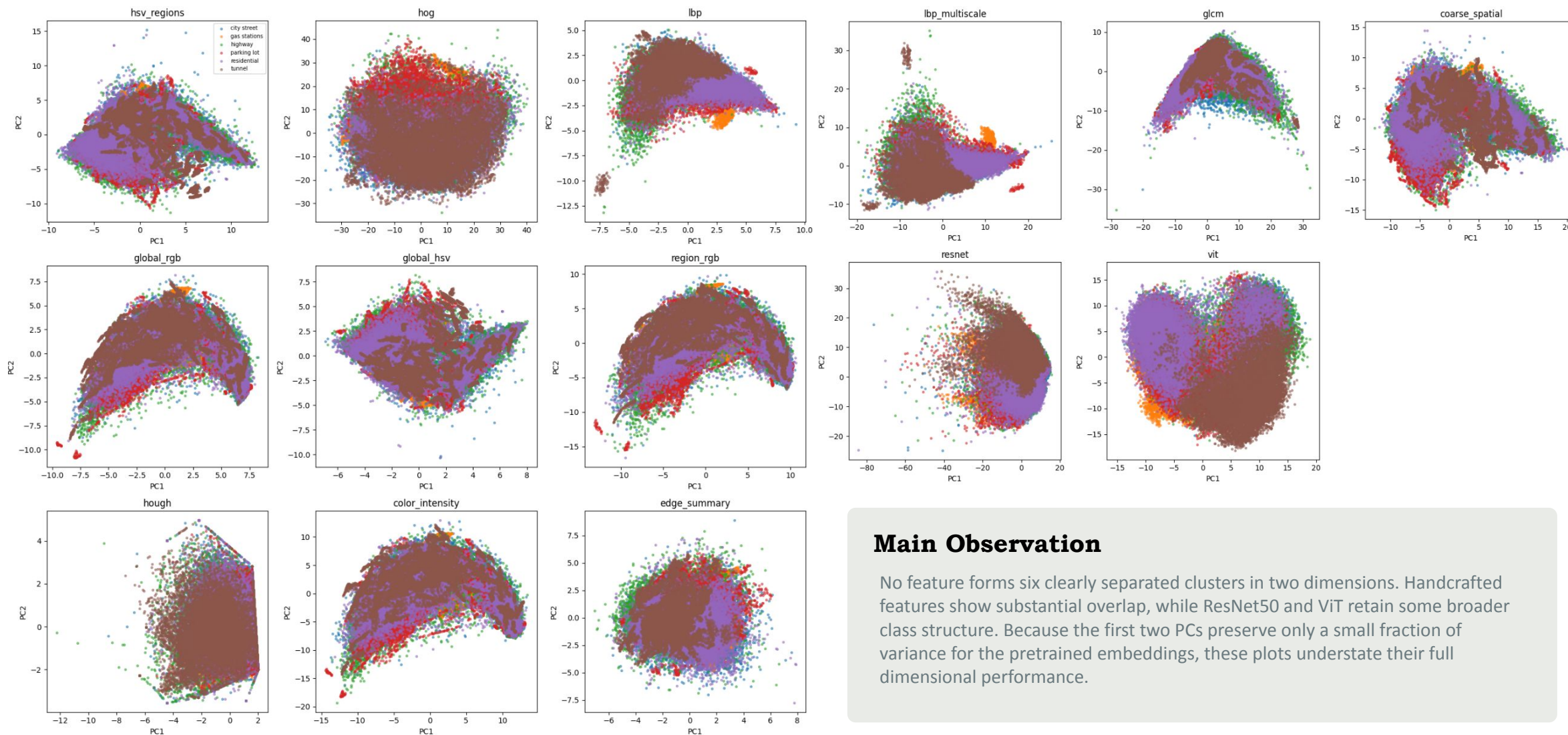


Reading the result

- GLCM and LBP are highly compressible: their first two components explain 90.35% and 79.14% of variance.
- ResNet50, HOG, ViT, and coarse spatial features spread variance across many components.
- Global color histograms compress more quickly than region-wise features, consistent with their loss of spatial detail.
- High explained variance does not necessarily mean better class separation or classification performance.

Feature	Variance in first 2 PCs
GLCM	90.35%
LBP	79.14%
Coarse spatial	17.57%
ViT-B/16	13.30%
HOG	7.00%
ResNet50	6.32%

Principal Component Analysis



Main Observation

No feature forms six clearly separated clusters in two dimensions. Handcrafted features show substantial overlap, while ResNet50 and ViT retain some broader class structure. Because the first two PCs preserve only a small fraction of variance for the pretrained embeddings, these plots understate their full dimensional performance.

Only the pretrained embeddings cluster by class

Stratified 2,000-image subset of the balanced training set, retaining every frame from the rarest classes. Each descriptor was standardised, reduced to 30 principal components, then embedded with perplexity 30.

Colour + intensity

Night frames of every class collapse together; the descriptor tracks illumination more than layout.

LBP / GLCM

Almost no class structure - texture alone cannot distinguish an urban facade from a suburban one.

Coarse spatial

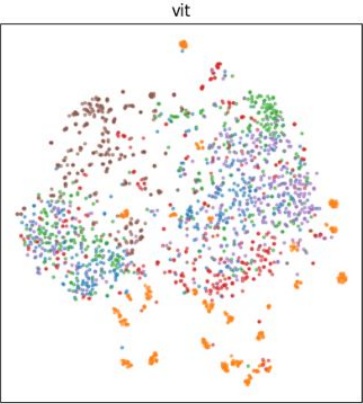
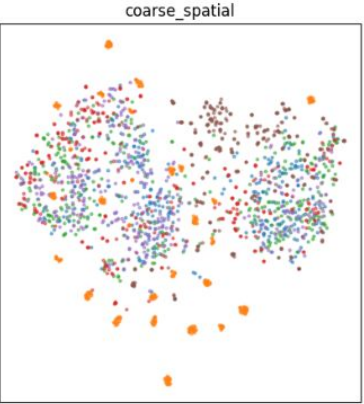
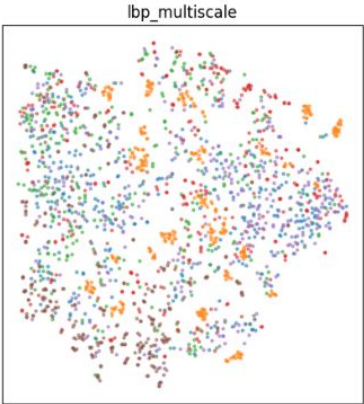
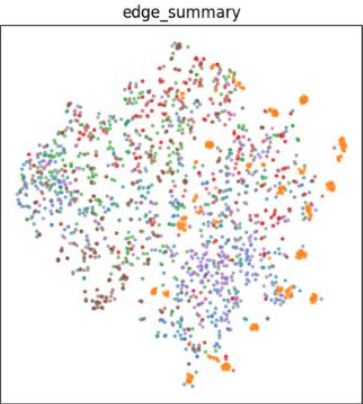
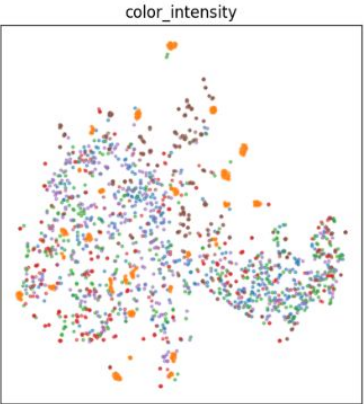
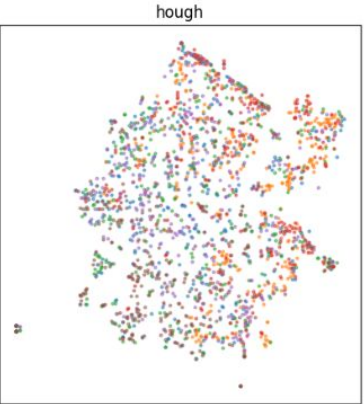
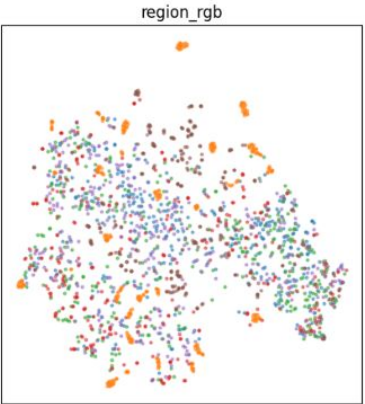
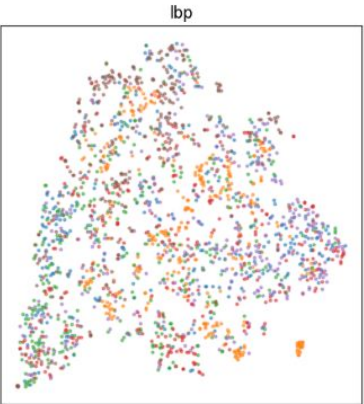
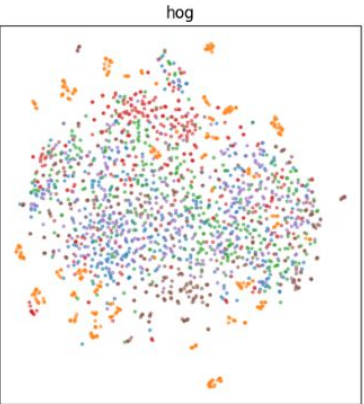
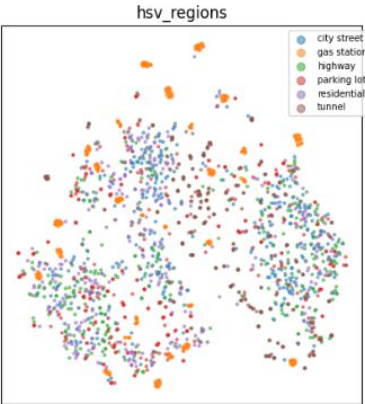
Partial separation of highway on its open sky-and-road layout, but city street and residential stay entangled.

ResNet50 / ViT

The only embeddings with visible class islands. Tunnel and highway form distinct groups; residential still overlaps city street.

Takeaway · The city street / residential overlap visible here is also shown in the final confusion matrix.

T-SNE VISUALISATION · 2 OF 2



Feature selection removed HOG and region HSV

Greedy leave-one-out elimination on a 15,000-image stratified subsample of train, scored by validation macro-F1 with logistic regression. A feature is dropped only if removing it improves macro-F1 by more than 0.002.

1	12 features	0.2818	baseline - all handcrafted descriptors
2	remove HOG	0.3361	+0.0543 - the single largest gain in the whole pipeline
3	remove region HSV	0.3464	+0.0103 - our proposal's designated primary colour feature
4	10 features	0.3464	Rejected removing other variables due to not meeting the macro-F1 threshold

What this tells us

- HOG's 4,032 dimensions swamped the other 1,100 in a shared linear model - its information was redundant with the cheaper edge and spatial descriptors
- Region-wise HSV was our proposal's headline colour feature; it turned out to duplicate the colour/intensity descriptor
- Dropping both cut handcrafted dimensionality from 5,132 to 956: an 81% reduction, while raising macro-F1
- Ten handcrafted descriptors survive, forming Set A:
 - lbp (10 dims), global_rgb (48 dims), global_hsv (48 dims), region_rgb (144 dims), hough (18 dims), color_intensity (176 dims), edge_summary (14 dims), lbp_multiscale (84 dims), glcm (72 dims), coarse_spatial (342 dims)

Split Protocol, Feature Sets, Hyperparameter Tuning

Split protocol

Train **69,996** resampled to 11,666 per class

Validation **9,947** official val split, natural imbalance

Test **19,902** official test split, touched only once

Splits are the dataset's own — no random re-partitioning, so there is no risk of the same clip appearing on both sides.

StandardScaler is fitted on the training set only and applied unchanged to validation and test, so no scaling statistics leak.

Feature sets and search

Set A — handcrafted only **956 dims**

The 10 descriptors that survived backward elimination.

Set B — handcrafted + complex **3,772 dims**

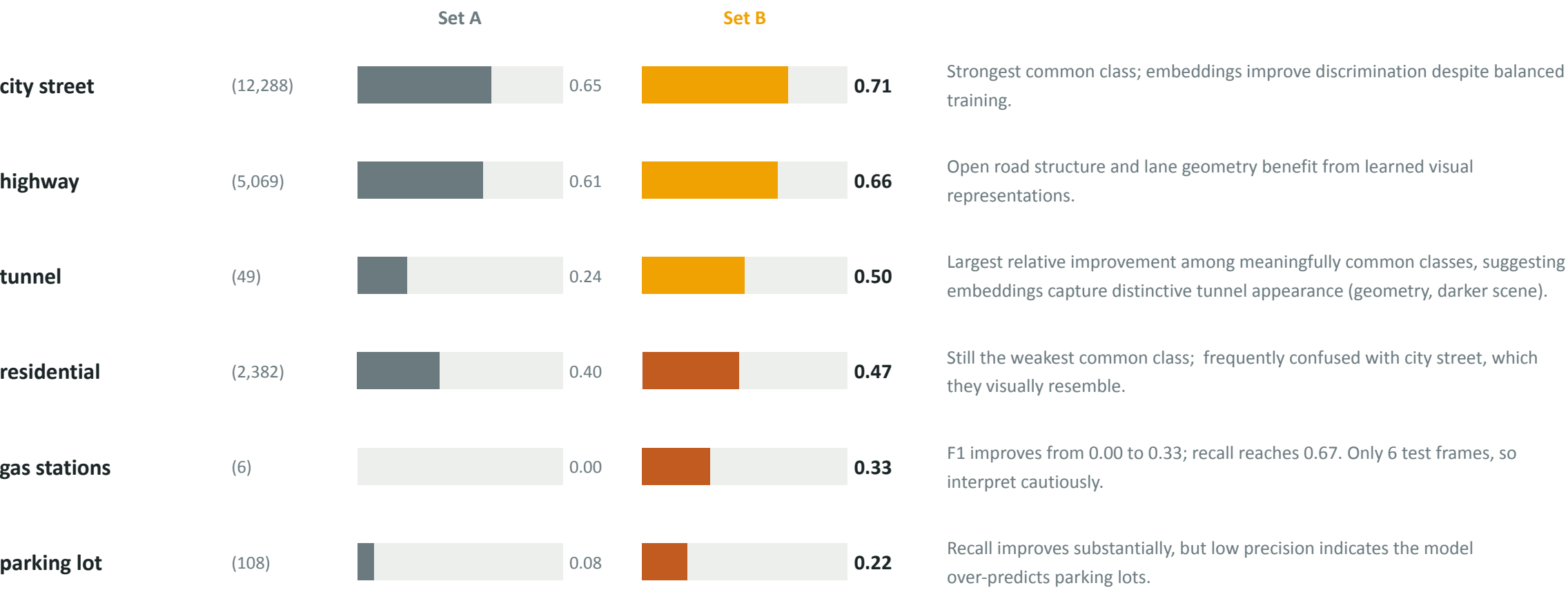
Set A concatenated with ResNet50 and ViT-B/16 embeddings.

Hyperparameter search

- RandomizedSearchCV over the inverse regularisation strength C , scored by stratified-CV macro-F1
- Winner selected on validation macro-F1, then a single held-out test evaluation
- Set A + LogReg: RandomizedSearchCV, 10 candidates $\times$ 3-fold CV, L2/lbfgs, C searched over $[0.01, 100]$; winner $C = 0.985$, CV macro-F1 = 0.770 $\rightarrow$ val macro-F1 = 0.353
- Set B + LogReg: RandomizedSearchCV, 4 candidates $\times$ 2-fold CV, L2/lbfgs, C searched over $[0.01, 10]$; winner $C = 0.133$, CV macro-F1 = 0.816 $\rightarrow$ val macro-F1 = 0.465

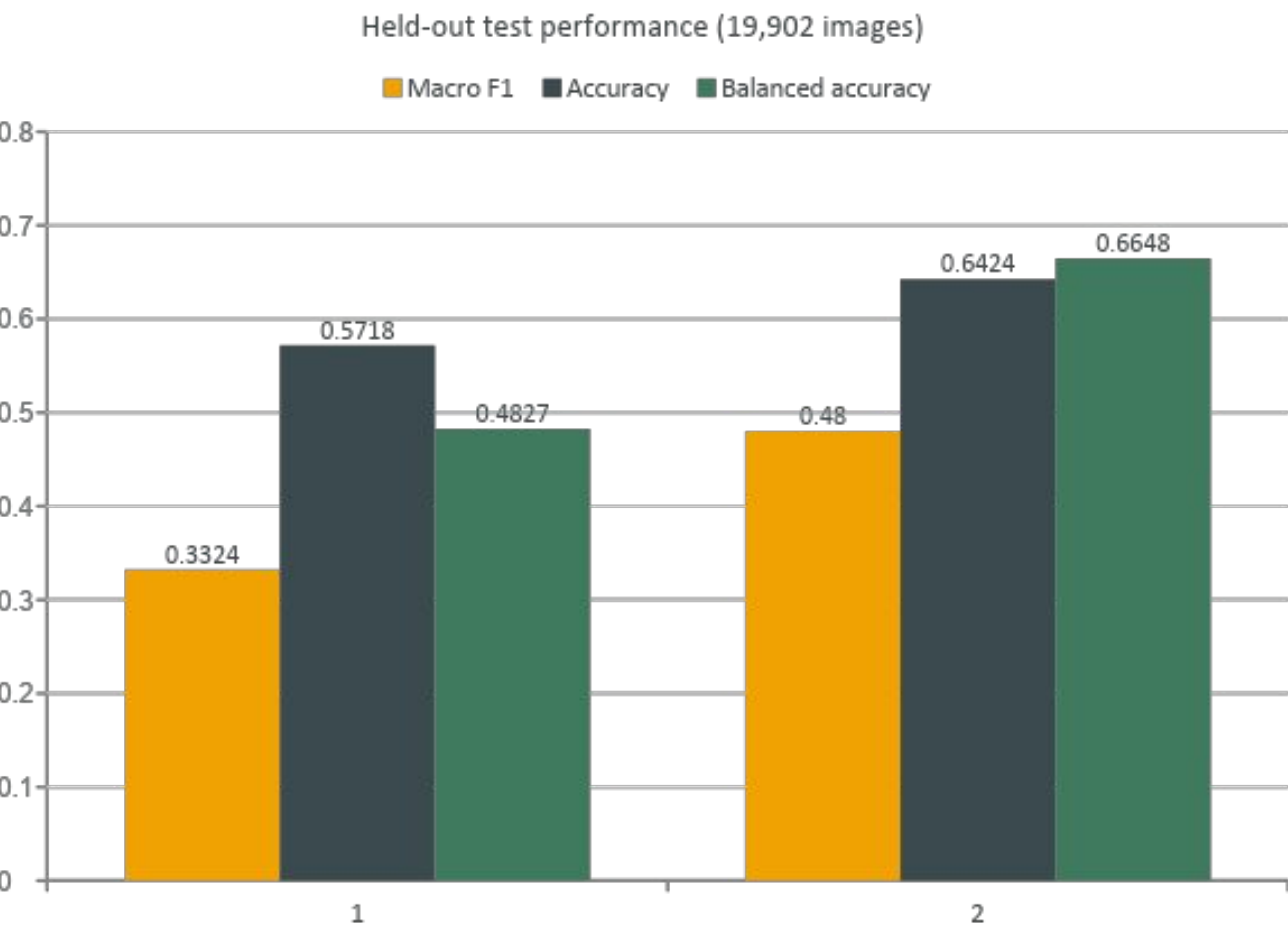
Learned embeddings improve every class

Per-class F1 on the held-out test set. Set A is handcrafted only; Set B adds ResNet50 and ViT embeddings.



Balanced accuracy rose from 0.483 to 0.665 between the two feature sets, and from 0.466 in our pre-resampling pipeline. The model is substantially less biased toward the majority classes, although performance on the rarest classes remains uncertain.

Classification Results by feature set (Set A vs Set B)



BEST MODEL - Set B LogReg

0.480 test macro-F1

Model comparison

Model	CV F1	Val F1	Test F1
Set A · LogReg	0.770	0.353	0.332
Set B · LogReg	0.816	0.465	0.480

Above is “Test” subset for Macro F1, Accuracy, and Balanced Accuracy for both Set A and Set B
Only Train is resampled data from resampled.zip (to achieve 16.7% share for each class).
Val and Test are untouched and straight from BDD100k archive (imbalanced class proportions e.g. 0.5% parking lot share)

Set B LogReg CVF1 - Val/Test gap



Some difficulty with generalization per training on rebalanced/duplicate (albeit augmented) images for rarer classes e.g. gas station

TWO LIKELY CAUSES OF GAP BETWEEN CROSS-VALIDATED AND HELD-OUT MACRO-F1

1 Different class balance

CV folds are drawn from the balanced training set, where macro-F1 rewards a classifier that is right about equally-sized classes. Validation and test are long-tailed, so the same model scores far lower. I.e. **rarer classes in Val/Test are over predicted by model training on rebalanced 16.7% class share data.**

2 Duplication inside the folds

Rare classes were oversampled up to 11,666 from a few hundred originals, so near-duplicate frames appear in both the fitting and held-out folds.

Train, Val, Test Accuracy for Set A and B

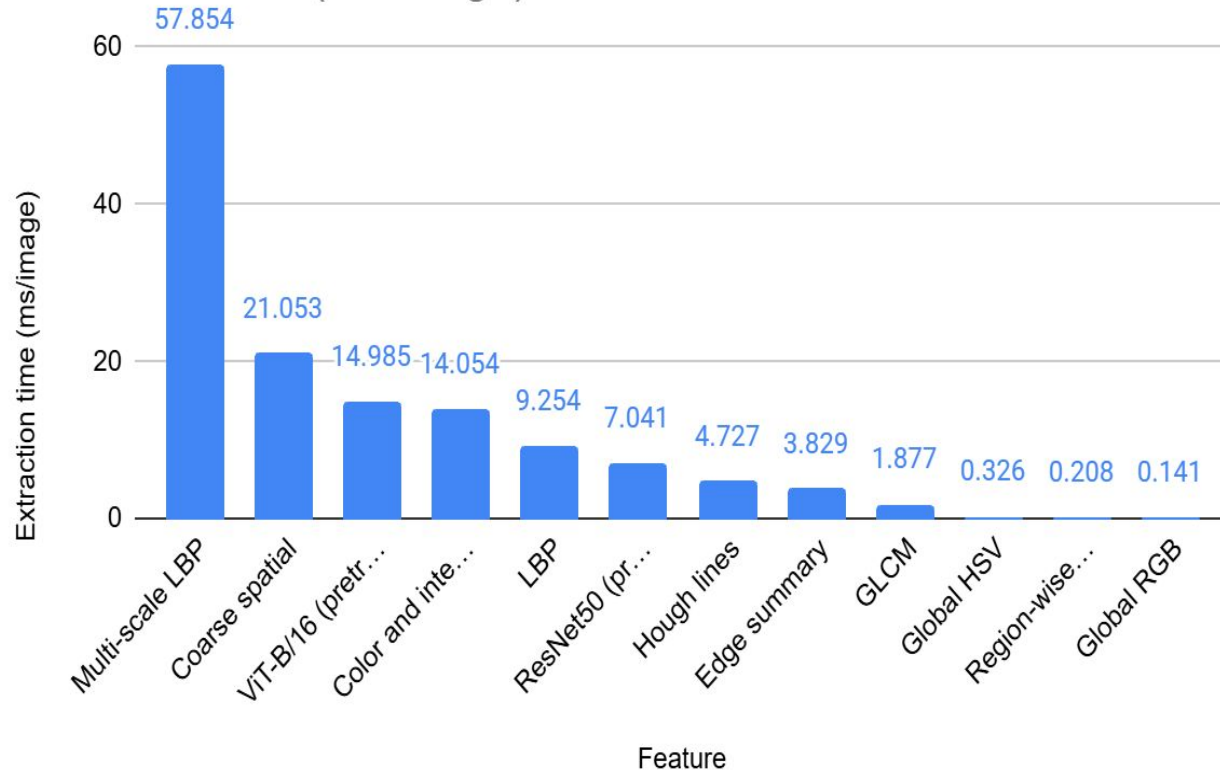
Balanced accuracy across all three splits				
	Train	Val	Test	Train-Test
Set A	0.790	0.534	0.483	-0.307
Set B	0.906	0.606	0.665	-0.241

High train accuracy likely result of re-balanced data image duplication, 54% of training set are duplicated (though augmented) images

class	original train	resampled to	factor	unique images
city street	43,581	11,666	0.3× down	11,666
highway	17,414	11,666	0.7× down	11,666
residential	8,105	11,666	1.4× up	8,105
parking lot	378	11,666	31× up	378
tunnel	129	11,666	90× up	129
gas stations	27	11,666	432× up	27

Per-feature extraction time in recorded Colab run

Extraction time (ms/image) vs. Feature



Accuracy-focused Representation

Set B + LogReg

0.480 test macro-F1 · 0.665 balanced accuracy · 135.3
ms/image · 3,772 dims · Pretrained weights required during
extraction

Efficiency-focused Representation

Set A + LogReg

0.332 test macro-F1 · 0.483 balanced accuracy · 113.3
ms/image · 956 dims · Handcrafted features only

Adding ResNet50 and ViT-B/16 increased estimated feature computation by 19.4% while improving test macro F1 by 0.1476. Multi-scale LBP was the slowest individual extractor at 57.9 ms/image—about 3.9× slower than ViT and 8.2× slower than ResNet50 in the recorded CPU run. Removing HOG and region-wise HSV saved approximately 6.6 ms/image and also improved validation macro F1.

Limitations

Limitations of the current classifier

- Frozen feature extractors
 - ResNet50 and ViT-B/16 were kept frozen and paired with logistic regression, so the model could not learn new task-specific visual features
- Ambiguous single-frame labels
 - Some categories overlap visually, especially city street and residential, and one keyframe removes potentially useful context
- Limited diversity from augmentation
 - Random shifts and crops balanced the training set, but related variants still came from the same original images
- Distribution mismatch
 - Training was balanced, while validation and test retained their imbalanced natural distributions

Rare Class table

Class	Test images	Precision	Recall
Gas stations	6	0.22	0.67
Tunnel	49	0.38	0.71
Parking lot	108	0.13	0.59

Recall change per test image

Class	Test images	Recall change per image
Gas stations	6	16.7 percentage points
Tunnel	49	2.0 percentage points
Parking lot	108	0.9 percentage points

Each value is 1/test images, so it shows how much recall changes if one additional image is classified correctly or incorrectly.